

HANA SENTINEL

Autonomous Policy-Driven Multi-Agent AI
for SAP HANA Operations

Product Requirements Document

Google x SAP Hackathon 2026

Version 2.0 | February 2026

CONFIDENTIAL

1. Vision Statement

An autonomous, explainable, policy-gated multi-agent system that transforms SAP HANA operations from reactive firefighting into proactive, self-healing infrastructure. Powered by Google Vertex AI, LangGraph orchestration, and a novel Risk Budget governance engine.

2. Problem Statement and Market Context

SAP HANA powers 87% of total global commerce through SAP's enterprise customer base. Yet the operational burden of maintaining HANA systems remains stubbornly manual. SAP Basis administrators perform daily routines that are well-defined, repetitive, and prime candidates for intelligent automation: checking system health, reviewing alerts, verifying backups, analyzing expensive SQL statements, monitoring capacity, and responding to security advisories.

The problem breaks into four dimensions. First, reactive operations: teams discover issues from EarlyWatch Alert reports (delivered weekly as PDFs) or after users report problems, not before impact occurs. Second, fragmented tooling: SAP Landscape Management (mainstream maintenance ending December 2027), SAP Automation Pilot (primarily HANA Cloud, not on-premise), and SAP Solution Manager (also end-of-life 2027) leave significant gaps. Third, knowledge silos: remediation expertise lives in the heads of senior Basis administrators, encoded in scattered SAP Notes and tribal knowledge rather than actionable automation. Fourth, compliance burden: SOX, GDPR, and now the EU AI Act demand auditable, explainable operations, making ad-hoc automation insufficient without governance frameworks.

Existing solutions like Avantra provide ML-based anomaly detection but stop short of autonomous remediation. SAP's own Joule copilot focuses on conversational assistance rather than operational autonomy. No solution today combines detection, diagnosis, explainable remediation, policy-gated execution, and continuous learning in a unified multi-agent architecture for SAP HANA.

Quantified Pain Points

Metric	Current State	Target with HANA Sentinel
MTTR for HANA incidents	~47 minutes	<8 minutes (83% reduction)
Time on routine Basis tasks	60-70% of FTE	<15% of FTE
EWA report review turnaround	2-5 business days	<1 hour (automated)
Backup failure detection	Next-morning check	Real-time (<30 seconds)
Security drift detection	Monthly/quarterly audit	Continuous (every 15 min)
Security patch assessment	Manual SAP Note review	Auto-parsed within 4 hours of Patch Day
Post-incident documentation	Manual, 30-60 min	Auto-generated X-Fix report

3. Real-World Threat Context: SAP Security Patch Day, January 2026

On January 13, 2026, SAP released 17 new security notes addressing critical vulnerabilities across enterprise SAP environments. This release included four HotNews vulnerabilities with CVSS ratings up to 9.9, four High priority issues, seven Medium priority fixes, and two Low priority updates. The patches affect SAP S/4HANA, SAP HANA database, SAP NetWeaver, SAP Wily Introscope, and various application components. This real-world event directly validates HANA Sentinel's Security Agent and Browser-Use Agent capabilities.

Critical Vulnerabilities (HotNews, CVSS 9.0+)

SAP Note	CVE	CVSS	Component	Vulnerability
3687749	CVE-2026-0501	9.9	S/4HANA General Ledger	SQL Injection via RFC function module using ADBC framework. Low-privilege users can execute arbitrary SQL, full system compromise possible.
3668679	CVE-2026-0500	9.6	Wily Introscope Enterprise Mgr	Remote Code Execution via malicious JNLP files. Unauthenticated attacker, user interaction required. Complete system takeover.
3694242	CVE-2026-0498	9.1	S/4HANA (Private Cloud/On-Prem)	Code Injection allowing admin-privilege attackers to modify program source code, leading to OS command execution.
3697979	CVE-2026-0491	9.1	Landscape Transformation (SLT)	Same code injection as above, shipped as separate DMIS add-on. Affects standalone SLT servers.

High Priority Vulnerabilities (CVSS 8.0+)

SAP Note	CVE	CVSS	Component	Vulnerability
3691059	CVE-2026-0492	8.8	SAP HANA Database	Privilege Escalation: authenticated users can switch to another user, potentially gaining admin access. Total compromise of confidentiality, integrity, availability.
3675151	CVE-2026-0507	8.4	AS ABAP / NW RFCSDK	OS Command Injection allowing high-privileged attackers on adjacent networks to execute arbitrary OS commands.
3688703	CVE-2026-0506	8.1	NW AS ABAP Platform	Missing Authorization Check enabling low-privilege users to bypass controls and perform unauthorized actions.

3565506	CVE-2026-0511	8.1	Fiori (Intercompany Balance)	Missing Authorization plus CSRF plus Information Disclosure across three CVEs.
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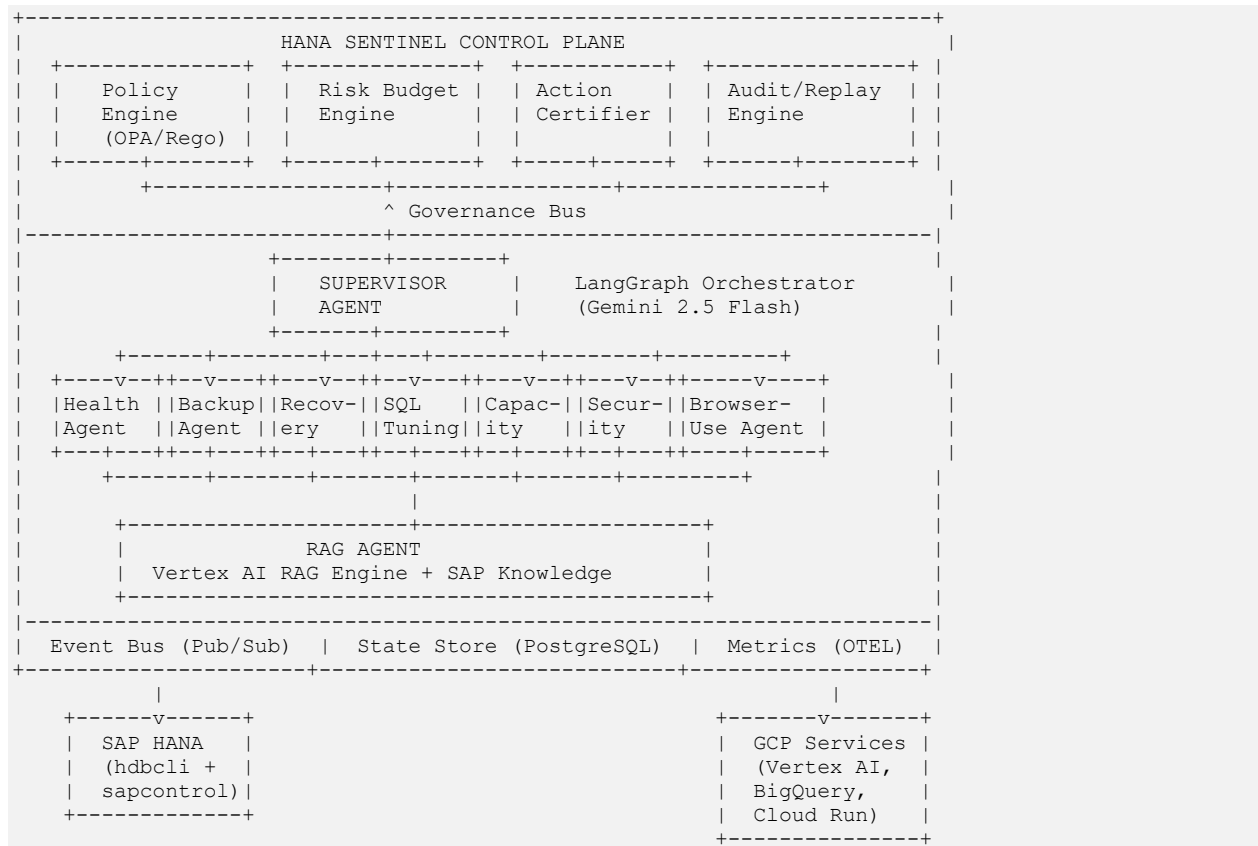
How HANA Sentinel Addresses January 2026 Patch Day

The January 2026 Patch Day demonstrates exactly the problem HANA Sentinel solves. Within hours of SAP publishing security notes, the system performs the following automated workflow:

- Browser-Use Agent navigates to SAP Support Portal and extracts the full list of security notes, CVE identifiers, affected components, and CVSS scores from the Patch Day bulletin page.
- RAG Agent ingests the extracted security notes into the Vertex AI knowledge corpus, making them immediately queryable by all operational agents.
- Security Agent cross-references affected components (especially SAP Note 3691059 for HANA privilege escalation CVE-2026-0492) against the managed HANA landscape, generating an Action Certificate for each applicable patch.
- For CVE-2026-0492 (HANA Privilege Escalation, CVSS 8.8), the Security Agent checks current HANA version, identifies exposed authentication boundaries, and generates an X-Fix report recommending immediate patching with blast radius analysis.
- Each remediation proposal flows through the Policy Engine and Risk Budget before reaching the Basis team for approval, with the X-Fix report providing full context for informed decision-making.

4. System Architecture and Multi-Agent Control Plane

Architecture Overview (ASCII)



Why LangGraph as the Orchestration Framework

LangGraph (GA May 2025, MIT-licensed, 400+ production deployments at companies like LinkedIn, Uber, and Elastic) models AI workflows as directed cyclic graphs with explicit state machines. This is critical for SAP operations because HANA remediation workflows are inherently stateful, branching, and require checkpointing.

- Reducer-driven state schemas using Python Annotated types prevent data loss in multi-agent environments
- Checkpointing via PostgresSaver enables time-travel debugging and fault-tolerant execution
- `interrupt()` function provides native human-in-the-loop support, pausing execution at policy gates
- Sub-graph composition allows each specialized agent to be its own LangGraph graph, composed into the supervisor's orchestration graph
- Native Vertex AI integration via `agent_engines.LanggraphAgent` class for managed deployment on Google Cloud

LangGraph State Schema

```
class HANAOperationState(TypedDict):
    messages: Annotated[list, add_messages]
    incident_id: str
    system_id: str
    risk_score: int
    risk_budget_remaining: int
    action_certificate: dict | None
    approval_status: Literal['pending', 'approved', 'rejected', 'auto']
    execution_status: Literal['idle', 'executing', 'completed', 'failed', 'rolled_back']
    xfix_explanation: dict | None
    audit_trail: list[dict]
```

Policy Engine (OPA/Rego)

The control plane uses Open Policy Agent (OPA), a CNCF-graduated project with sub-millisecond policy evaluation, as its decision engine. Policies are written in Rego and enforced before every agent action. The engine evaluates risk tiers (low/medium/high/critical), checks budget sufficiency, enforces maintenance windows for critical actions, and prevents agents from approving their own actions (separation of duties).

5. Agent Risk Budget (Novel Governance Mechanism)

HANA Sentinel introduces the Agent Risk Budget, adapted from Google's SRE error budget model. Just as SRE teams have an error budget (1 minus SLO) that permits controlled risk-taking while protecting reliability, each HANA Sentinel deployment operates with a risk budget that governs cumulative autonomous action.

Every agent action consumes risk points from a shared budget based on a 5x5 risk matrix (Likelihood x Impact). The budget replenishes over a rolling 24-hour window, with trust multipliers rewarding successful operations and penalty mechanisms reducing capacity after incidents.

Risk Scoring for SAP HANA Operations

Operation Category	Likelihood	Impact	Risk Points	Example
Read-only monitoring	1 (Rare)	1 (Negligible)	1	Query M_SERVICES status
Log/trace analysis	1	2 (Minor)	2	Parse expensive statement trace
Configuration read	1	1	1	Read M_INIFILE_CONTENTS
Parameter change	2 (Unlikely)	3 (Moderate)	6	Adjust trace level
global.ini modification	2	3 (Moderate)	6	Add basepath_logbackup entry
Backup execution	2	3	6	Trigger differential backup
Service restart	3 (Moderate)	4 (Major)	12	Restart indexserver
Security change	2	4 (Major)	8	Revoke excessive privilege
Data modification	3	5 (Severe)	15	Table reorganization
Failover trigger	4 (Likely)	5 (Severe)	20	System replication takeover

Budget Thresholds and Behavioral Gates

Budget Consumed	Agent Behavior	Governance Mode
0-50%	Fully autonomous for pre-approved actions	HOOTL (Human-out-of-the-loop)
50-75%	Autonomous with enhanced logging and explanation	HOTL (Human-on-the-loop)
75-90%	All actions require async human review	HITL-async
90-100%	All actions require synchronous human approval	HITL-sync
>100%	Read-only mode; only monitoring permitted	Frozen

Trust multiplier: Seven consecutive days without incidents increases the daily budget by 10% (compounding, capped at 150% of baseline). A severity-1 incident resets the multiplier and reduces budget to 75% of baseline for 48 hours.

6. Action Certificates (Signed Pre-Approved Remediation Plans)

Every non-trivial agent action in HANA Sentinel requires an Action Certificate, a digitally signed document that fully describes what will be done, why, what could go wrong, and how to undo it. This concept draws from Terraform's plan/apply model, AWS Systems Manager runbooks, and Columbia University's research on Autonomy Certificates (Feng et al., 2025).

Action Certificates serve three purposes: they enforce pre-execution review, they create an immutable audit record for compliance, and they enable the replay capability by fully capturing decision context.

Certificate Structure

Field Group	Fields	Purpose
Identity	certificate_id, version, status, created_by_agent, digital_signature, expires_at	Tracking, authentication, lifecycle
Target	system_id, target_component	What system and component is affected
Action	action_type, action_description, action_parameters, execution_steps	Exactly what will be done
Justification (WHY)	trigger_event, root_cause_hypothesis, confidence_level, alternatives_considered, supporting_evidence, rag_sources	Evidence-based reasoning chain
Risk Assessment	risk_score (1-25), likelihood, impact, blast_radius, risk_budget_cost	Quantified risk and affected scope
Pre/Post Conditions	preconditions[], postconditions[]	Validation before and after execution
Rollback	rollback_steps, auto_rollback_triggers, estimated_rollback_time, data_loss_risk	Full reversal plan
Approval	approval_required, approved_by, approved_at, approval_notes	Human governance record

7. Specialized Agent Specifications

Each agent follows the single-responsibility principle: one agent, one domain, clear input/output contracts. Agents communicate through structured intents over Google Cloud Pub/Sub (not natural language) to reduce hallucination risk and latency.

7.1 Health Monitor Agent

Continuous monitoring of SAP HANA system health via M_* monitoring views. Generates structured health assessments every 60 seconds covering services, memory, disk, CPU, alerts, and system replication.

Key HANA interfaces: M_SERVICES, M_HOST_RESOURCE_UTILIZATION, M_SERVICE_MEMORY, STATISTICS_CURRENT_ALERTS, M_DISK_USAGE, M_SYSTEM_REPLICATION, M_LOAD_HISTORY_SERVICE

7.2 Backup Agent

Manages the entire backup lifecycle: scheduling, execution, verification, catalog housekeeping, and failure recovery. Interfaces directly with HANA via hdbcli for BACKUP DATA commands and M_BACKUP_CATALOG monitoring.

Risk levels: Read catalog = 1 point; trigger differential = 6 points; trigger full = 6 points; catalog delete = 8 points.

7.3 Recovery Agent

Detects service failures and executes recovery procedures, from simple service restarts to system replication takeover. Monitors M_SERVICES for status changes and executes recovery via sapcontrol (OS-level commands via SSH) and SQL ALTER SYSTEM START SERVICE.

7.4 SQL Tuning Agent

Identifies expensive SQL statements, analyzes execution plans, and recommends or executes optimizations. Queries M_SQL_PLAN_CACHE, M_EXPENSIVE_STATEMENTS, and M_ACTIVE_STATEMENTS. Before recommending tuning actions, the SQL Tuning Agent queries the RAG Agent with the problematic SQL pattern to retrieve relevant SAP Notes and known performance issues.

7.5 Capacity Agent

Tracks disk, memory, and table growth trends. Predicts capacity thresholds and recommends actions before exhaustion. Integrates with the global.ini configuration management workflow (see Section 8) to ensure backup and data volume paths are correctly configured.

7.6 Security Agent

Continuous security posture assessment covering privilege auditing, password policy compliance, encryption verification, audit policy checks, and SAP Security Patch Day vulnerability assessment. Queries EFFECTIVE_PRIVILEGES, M_PASSWORD_POLICY, M_ENCRYPTION_OVERVIEW, and AUDIT_POLICIES.

For the January 2026 Patch Day specifically, the Security Agent would flag CVE-2026-0492 (HANA Privilege Escalation, CVSS 8.8) as directly affecting the managed HANA instance and generate an Action Certificate recommending immediate assessment and patching.

Agent Summary Table

Agent	Responsibility	Auto-Fix Scope	Risk Tier
Health Agent	KPIs, EWA findings, M_* views	Read-only (monitoring)	Low
Backup Agent	Backup compliance, catalog mgmt	Differential + full backup	Medium
Recovery Agent	Service failure recovery	Restart (approval for failover)	High
SQL Tuning Agent	Expensive SQL optimization	Index creation (approval for hints)	Medium
Capacity Agent	Disk/memory trending, global.ini	Trace cleanup, config updates	Medium
Security Agent	Parameter drift, CVE assessment	Privilege revocation (approval req)	High
RAG Agent	SAP knowledge retrieval + support	Read-only (advisory)	Low
Browser-Use Agent	SAP web UI automation	Navigation + extraction	Medium

8. Configuration Agent: global.ini Management

A critical operational requirement for HANA Express and production HANA systems is ensuring the global.ini configuration file contains correct backup and volume path entries. HANA Sentinel's Capacity Agent and Backup Agent jointly manage this configuration as part of their initialization and continuous compliance checks.

Target Configuration

File path: /usr/sap/HXE/SYS/global/hdb/custom/config/global.ini

The agents verify that the following entries exist under the [persistence] section:

```
[persistence]
basepath_databackup = /hdb/HXE/backup/data
basepath_logbackup = /hdb/HXE/backup/log
basepath_datavolumes = /hdb/HXE/data
basepath_logvolumes = /hdb/HXE/log
```

Agent Workflow for global.ini Compliance

Step 1 (Detection): On startup and every 15 minutes, the Capacity Agent reads the global.ini file via SSH and parses the [persistence] section.

Step 2 (Validation): The agent checks whether all four required entries (basepath_databackup, basepath_logbackup, basepath_datavolumes, basepath_logvolumes) are present with correct values.

Step 3 (Remediation): If any entries are missing, the agent generates an Action Certificate describing the exact changes needed. Since this is a parameter change (risk score: 6), it flows through the Policy Engine for approval in accordance with the current governance mode.

Step 4 (Execution): Upon approval, the agent appends the missing entries to global.ini using sed or direct file manipulation via SSH. It then verifies the changes by re-reading the file.

Step 5 (Verification): The agent queries M_INIFILE_CONTENTS via SQL to confirm HANA has picked up the configuration changes. If HANA requires a restart to apply changes, the agent escalates to the Recovery Agent with appropriate risk scoring.

Implementation Detail

```
# Agent checks global.ini via SSH
def check_global_ini(ssh_client):
    ini_path = '/usr/sap/HXE/SYS/global/hdb/custom/config/global.ini'
    required_entries = {
        'basepath_databackup': '/hdb/HXE/backup/data',
        'basepath_logbackup': '/hdb/HXE/backup/log',
        'basepath_datavolumes': '/hdb/HXE/data',
        'basepath_logvolumes': '/hdb/HXE/log'
    }

    stdin, stdout, stderr = ssh_client.exec_command(f'cat {ini_path}')
    content = stdout.read().decode()
```

```
missing = {}
for key, value in required_entries.items():
    if f'{key} = {value}' not in content:
        missing[key] = value

if missing:
    return generate_action_certificate(
        action_type='config_update',
        target='global.ini [persistence]',
        changes=missing,
        risk_score=6,
        rollback='Remove added entries from global.ini'
    )
return None # All entries present
```

The `basepath_logbackup` entry is particularly critical because it enables automatic log backups, which are essential for point-in-time recovery. Without this entry, HANA's log area can fill up and block all transactions. The Backup Agent depends on this path being correctly configured before it can execute any log backup operations.

9. RAG Agent: Vertex AI-Powered Knowledge Engine

The RAG Agent serves dual roles: it grounds operational agents' decisions in SAP documentation (ensuring recommendations are based on official SAP Notes, not LLM hallucinations), and it powers a user-facing support assistant that Basis administrators can query in natural language.

Ingestion Pipeline

- SAP EWA Reports (PDF) parsed via Document AI Layout Parser
- SAP Notes (HTML/PDF) from SAP Support Portal
- HANA Administration Guides (PDF)
- SAP KBAs (Knowledge Base Articles)
- SAP Security Patch Day bulletins (HTML, extracted by Browser-Use Agent)
- Internal runbooks and operational procedures

Vertex AI RAG Engine Configuration

```
from vertexai import rag
import vertexai

vertexai.init(project='hana-sentinel', location='us-central1')

sap_corpus = rag.create_corpus(
    display_name='sap-hana-knowledge-base',
    description='SAP Notes, EWA reports, HANA admin guides, Patch Day bulletins'
)

rag.import_files(
    corpus_name=sap_corpus.name,
    paths=[
        'gs://hana-sentinel-docs/sap-notes/',
        'gs://hana-sentinel-docs/ewa-reports/',
        'gs://hana-sentinel-docs/hana-admin-guides/',
        'gs://hana-sentinel-docs/patch-day-bulletins/'
    ],
    transformation_config=rag.TransformationConfig(
        rag.ChunkingConfig(chunk_size=512, chunk_overlap=100)
    ),
    max_embedding_requests_per_min=900
)
```

Embedding model: gemini-embedding-001 (3072 dimensions, 8,192 token input limit). Hybrid search (semantic + keyword) is critical because keyword search catches exact SAP Note numbers (e.g., SAP Note 2222200) and error codes, while semantic search handles conceptual queries.

Dual-Role Operation

Role 1 (Agent Grounding): When operational agents need to make decisions, they query the RAG Agent to retrieve relevant documentation. The SQL Tuning Agent, for example, queries the RAG corpus with a problematic SQL pattern before recommending optimization. The response

includes citations to specific SAP Notes and admin guide sections, which are embedded into the Action Certificate's rag_sources field.

Role 2 (Support Assistant): Basis administrators can query the RAG Agent directly via the HANA Sentinel UI or CLI. Example: 'How do I reduce memory pressure in HANA columnstore?' returns an answer grounded in SAP Note 2222200 and the HANA Administration Guide Chapter 12, with confidence scores and source citations.

10. Browser-Use Agent: SAP Web Interface Automation

The Browser-Use Agent leverages the browser-use Python framework (v0.10.1) to automate interactions with SAP web interfaces that lack programmatic APIs. Built on Chrome DevTools Protocol (CDP), it uses a hybrid DOM analysis + visual understanding approach where an LLM determines actions based on both the serialized DOM tree and screenshots, making it resilient to SAP Fiori's notoriously dynamic element IDs.

Key Use Cases

- SAP Security Patch Day monitoring: Navigate to support.sap.com, extract security note lists, CVE identifiers, CVSS scores, and affected components from the monthly Patch Day bulletin.
- HANA Cockpit data extraction: Navigate to System Overview tiles, extract health dashboards, capture KPI visualizations as audit screenshots.
- EWA report retrieval: Access Solution Manager, navigate to EarlyWatch Alert Management, extract the latest report's Service Summary, Performance, and Security sections.
- SAP Note detail extraction: For each relevant security note (like 3691059 for HANA privilege escalation), extract full remediation steps, affected versions, and workaround instructions.
- Configuration operations: Perform administrative tasks in HANA Cockpit that are not exposed via SQL.

Security Configuration

```
sap_profile = BrowserProfile(  
    headless=True,  
    viewport={'width': 1920, 'height': 1080},  
    wait_for_network_idle_page_load_time=5.0, # SAP Fiori is AJAX-heavy  
    minimum_wait_page_load_time=3.0,  
    wait_between_actions=2.0,  
    allowed_domains=[                                # PREVENTS PROMPT INJECTION  
        'support.sap.com',  
        'me.sap.com',  
        'sap-hana-cockpit.internal.company.com',  
        'solman.internal.company.com'  
    ],  
    storage_state='sap_auth_state.json',             # Persist auth cookies  
    device_scale_factor=2.0                           # High-res audit screenshots  
)
```

Security is non-negotiable: `sensitive_data` must always be paired with `allowed_domains`. Without domain restrictions, prompt injection attacks on visited pages could expose credentials. All sessions use `storage_state` persistence to minimize re-authentication, and every step triggers an audit screenshot hook.

11. Explainable Remediation (X-Fix)

Every automated fix generates an X-Fix report, a structured, human-readable explanation that maps directly to ITIL Change Management RFC fields. Critical design principle: X-Fix explanations are grounded in observable system state, not LLM-generated narratives. Every claim in an X-Fix report links to specific metrics, logs, or trace data.

X-Fix Report Structure

Section	Content	Purpose
SUMMARY	One-paragraph description of issue and proposed fix	Quick executive overview
WHY (Root Cause)	Trigger event, evidence chain with metric links, confidence score, RAG sources, alternatives considered	Auditable reasoning chain
WHAT (Proposed Actions)	Numbered steps with commands, estimated duration per step	Exact execution plan
RISK (Impact Assessment)	Risk score (1-25), blast radius, affected processes, budget cost, approval requirement	Informed risk decision
ROLLBACK	Auto-trigger conditions, reversal steps, estimated time, data loss risk	Safety net
EXPECTED OUTCOME	Target metrics post-fix, verification method, escalation criteria	Success criteria

12. Replayable Incidents and Continuous Learning

HANA Sentinel records every incident with full system state, event streams, and agent decision traces, enabling sandbox replay for continuous improvement. This is the system's learning flywheel: incidents become training data, what-if scenarios validate policy changes, and historical pattern matching (via vector similarity search on pgvector embeddings) accelerates future diagnosis.

13. Lab Environment: GCP + HANA Express Setup

The hackathon development and demonstration environment runs on Google Cloud Platform with SAP HANA Express Edition in Docker. This section documents the exact infrastructure configuration for reproducibility.

GCP Compute Instance

Parameter	Value
Instance name	instance-20260211-142542
Machine type	n2-standard-8 (8 vCPU, 32 GB RAM)
CPU platform	Intel Cascade Lake (2 threads/core)
Zone	us-central1-a
Project	ai-connect-sap26blr-325
OS image	SLES SAP 15 (SUSE Linux Enterprise Server for SAP)
Boot disk	40 GB Persistent (SCSI)
Data disk	100 GB Persistent (SCSI, /dev/sdb, mounted at /data)
External IP	35.238.233.131
Internal IP	10.128.0.2
Network tier	PREMIUM
Shielded VM	Integrity monitoring ON, vTPM ON, Secure Boot OFF
Status	RUNNING

HANA Express Docker Configuration

```
sudo docker run -d \  
  -p 39013:39013 \  
  -p 39015:39015 \  
  -p 39041-39045:39041-39045 \  
  -h hxehost \  
  -v /data/hxe:/hana/mounts \  
  --ulimit nofile=1048576:1048576 \  
  --sysctl kernel.shmmax=1073741824 \  
  --name hxehana \  
  saplabs/hanaexpress:latest \  
  --passwords-url file:///hana/mounts/hxe_password.json \  
  --agree-to-sap-license
```

Port Mapping

Port	Service	Purpose
39013	HANA SQL/MDX (System DB)	Primary database connection via hdbcli
39015	HANA SQL/MDX (Tenant HXE)	Tenant database connection
39041	HANA Cockpit	Web-based administration UI (Browser-Use Agent target)

39042	HANA XSA Controller	Extended Application Services
39043-39045	HANA XSA Runtime	Application runtime ports

Connection Parameters for Agents

```
# Python hdbcli connection (used by all operational agents)
import hdbcli.dbapi

conn = hdbcli.dbapi.connect(
    address='35.238.233.131',      # External IP for dev; 10.128.0.2 for prod
    port=39013,                  # System DB port
    user='SYSTEM',
    password=vault.get_secret('hana/hxe/system_password'), # HXEHana1 in dev
    encrypt=True,
    sslValidateCertificate=False  # Dev only; enforce in production
)

# SSH connection for sapcontrol and global.ini management
ssh_config = {
    'hostname': '35.238.233.131',
    'username': 'hxeadm',
    'key_filename': '/path/to/gcp_ssh_key',
    'port': 22
}
```

Note: The master password (HXEHana1) is used exclusively in the development environment. Production deployments must use Secret Manager with automatic rotation and IAM-scoped access for each agent's service account.

Docker Volume Mapping

Host path `/data/hxe` maps to container path `/hana/mounts`. The password configuration file at `/hana/mounts/hxe_password.json` contains the master password used during HANA Express initialization. The data disk (100 GB, `/dev/sdb`) provides dedicated I/O for HANA data and log volumes, separate from the OS boot disk.

14. REST API Contracts

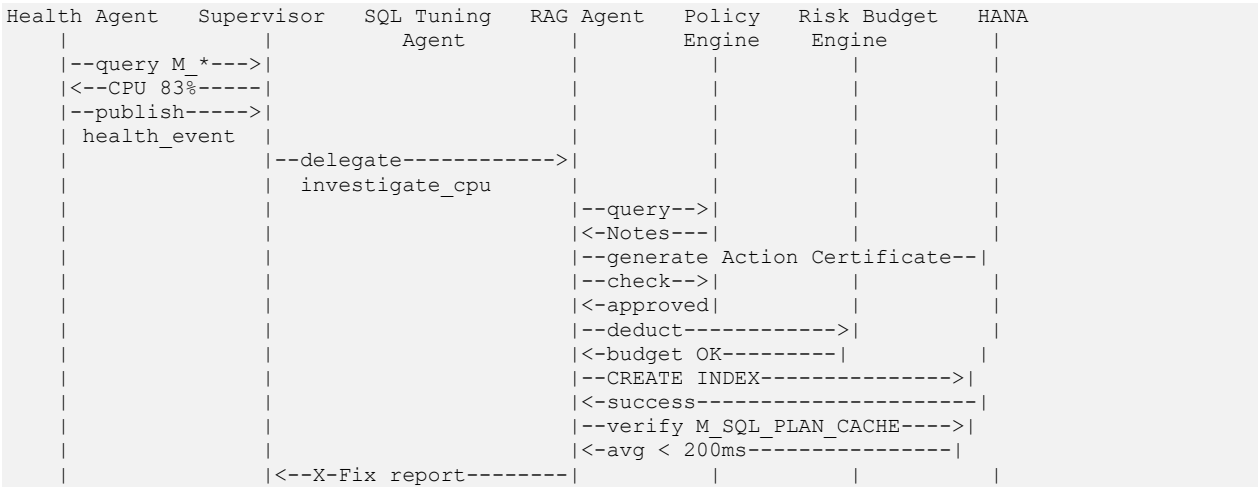
The Control Plane exposes a FastAPI-based REST API for all operations. All endpoints follow a consistent envelope pattern with OpenTelemetry trace correlation.

Core API Endpoints

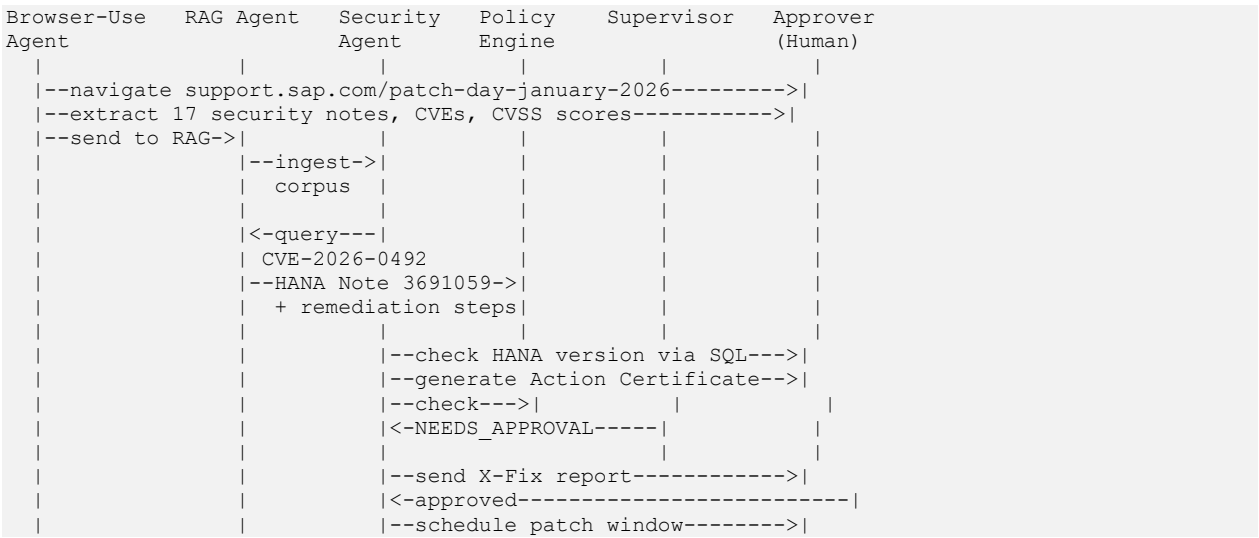
Method	Endpoint	Description
POST	/api/v1/incidents	Create incident from agent detection
GET	/api/v1/incidents/{id}	Retrieve incident details
GET	/api/v1/incidents/{id}/timeline	Full event timeline
POST	/api/v1/incidents/{id}/remediation	Propose X-Fix remediation
POST	/api/v1/remediations/{cert_id}/approve	Human approval of Action Certificate
POST	/api/v1/remediations/{cert_id}/execute	Trigger execution
POST	/api/v1/remediations/{cert_id}/rollback	Trigger rollback
GET	/api/v1/risk-budgets/{system_id}	Current risk budget status
GET	/api/v1/risk-budgets/{system_id}/transactions	Budget transaction history
GET	/api/v1/agents	List all registered agents
GET	/api/v1/agents/{id}/decisions	Agent decision history
POST	/api/v1/replays	Create incident replay
GET	/api/v1/replays/{id}/comparison	Original vs replay comparison
POST	/api/v1/rag/query	Query RAG knowledge base
POST	/api/v1/config/global-ini/check	Validate global.ini entries
POST	/api/v1/security/patch-day/assess	Assess Patch Day impact on landscape

15. Sequence Diagrams for Key Flows

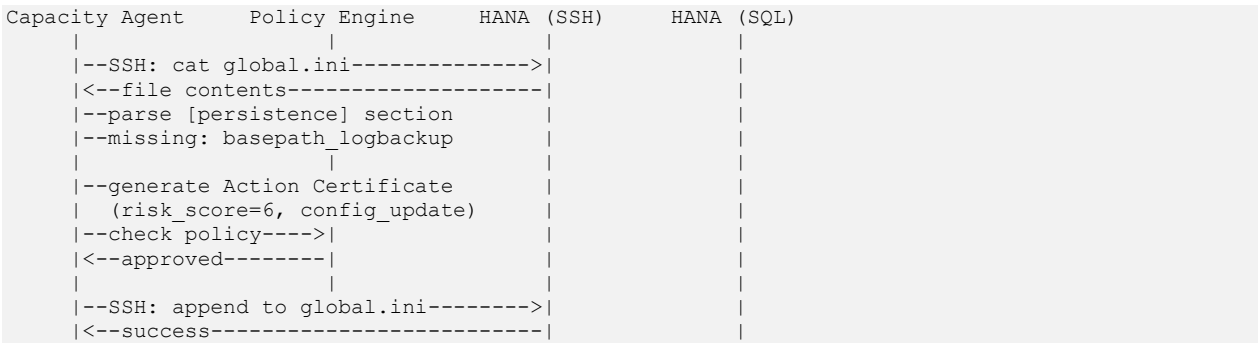
Flow 1: Issue Detection to Autonomous Remediation



Flow 2: Security Patch Day Assessment (January 2026)



Flow 3: global.ini Configuration Compliance



```
|--SQL: SELECT FROM M_INIFILE_CONTENTS----->|
|<--verified: entries present-----|
|--publish config_compliant event      |      |
```

16. Complete Technology Stack

Layer	Technology	Purpose
Orchestration	LangGraph on Vertex AI Agent Engine	Multi-agent state machines, HITL, checkpointing
LLM (Primary)	Gemini 2.5 Flash	Agent reasoning, X-Fix generation (1M context)
LLM (Complex)	Gemini 2.5 Pro	Deep root cause analysis, complex SQL tuning
RAG Engine	Vertex AI RAG Engine	Doc ingestion, hybrid search, grounded generation
Embeddings	gemini-embedding-001	3072-dim vectors for RAG and incident similarity
API Layer	FastAPI on Cloud Run	REST API control plane (auto-scaling)
Event Bus	Google Cloud Pub/Sub	Inter-agent communication, event distribution
Policy Engine	Open Policy Agent (OPA)	Policy-as-code enforcement (Rego, sub-ms eval)
State Store	PostgreSQL (Cloud SQL)	Agent state, Action Certs, Risk Budgets, incidents
Vector Store	pgvector extension	Incident similarity search for replay engine
Audit Analytics	BigQuery	Long-term audit log storage and analytics
Object Storage	Cloud Storage	SAP documents, screenshots, agent artifacts
Observability	OpenTelemetry + Cloud Trace	Distributed tracing, metrics, alerting
Browser Automation	browser-use v0.10.1	SAP web UI interaction (CDP-based)
HANA Connectivity	Python hdbcli	Direct SQL execution (PEP 249 compliant)
OS Operations	SSH (paramiko) + sapcontrol	Service lifecycle, global.ini management
Dev/Test HANA	HANA Express (Docker)	saplabs/hanaexpress:latest on n2-standard-8
GCP Project	ai-connect-sap26blr-325	Hackathon project with Vertex AI APIs enabled
Secrets	Secret Manager	Credential storage with automatic rotation
CI/CD	Cloud Build	Container build and deployment

17. Chaos Engineering Test Suite

HANA Sentinel includes a built-in chaos engineering framework to validate agent responses in the HANA Express lab environment.

Scenario	Simulation Method	Expected Agent Response	Pass Criteria
Indexserver crash	kill -9 on hdbindexserver PID	Recovery Agent: detect, Action Cert, restart	Service GREEN within 120s
Log disk full	dd if=/dev/zero of=/hana/log/fill	Capacity Agent: detect, clean traces, alert	Disk usage <90% within 60s
Backup failure	Block backup target via iptables	Backup Agent: detect, retry alternate, alert	Backup completes within 300s
Memory pressure	Load large dataset into memory	Health + SQL Tuning: identify offender, unload	Memory <85% within 300s
Missing global.ini	Remove basepath_logbackup entry	Capacity Agent: detect, generate cert, restore	Entry restored within 90s
Security drift	Grant SAP_ALL to test user	Security Agent: detect, revoke, audit report	Privilege revoked within 60s
Patch Day alert	Simulate new CVE for HANA	Browser-Use + Security: extract, assess, report	Assessment generated <10min

18. Innovation Highlights for Hackathon Judges

Innovation 1: Agent Risk Budgets. First application of SRE error budget principles to autonomous AI agent governance. Creates a quantitative trust framework that enables graduated autonomy without binary 'fully manual' or 'fully automated' choices. No existing SAP automation tool implements this.

Innovation 2: Action Certificates. Digitally signed, pre-validated remediation plans with preconditions, postconditions, blast radius analysis, and automatic rollback triggers. Bridges the gap between infrastructure-as-code approval patterns (Terraform plan/apply) and AI agent autonomy.

Innovation 3: Explainable Remediation (X-Fix). Every automated action generates a structured explanation grounded in observable system state (not LLM narrative), mapping directly to ITIL RFC fields. This makes AI actions auditable for SOX, GDPR, and EU AI Act compliance.

Innovation 4: Incident Replay as a Learning Flywheel. Recorded incidents become training data for what-if analysis, enabling continuous agent improvement. Combined with vector-similarity search, new incidents are matched to historical patterns for accelerated diagnosis.

Innovation 5: Browser-Use + SAP Fiori. Novel application of LLM-driven browser automation to SAP's web interfaces. The hybrid DOM + vision approach handles Fiori's dynamic IDs naturally, eliminating brittle selector-based automation. Applied to SAP Security Patch Day monitoring for real-time vulnerability assessment.

Innovation 6: Chaos Engineering for SAP. Built-in failure simulation framework validates agent responses before production deployment. Very few teams apply chaos engineering principles to SAP infrastructure.

Google + SAP Partnership Alignment

- Uses Agent2Agent (A2A) protocol patterns (SAP is a founding contributor)
- Deploys on Vertex AI Agent Engine with native LangGraph support
- Leverages Gemini models available via SAP BTP Generative AI Hub
- Demonstrates zero-copy data patterns mirroring SAP BDC Connect for BigQuery
- Implements OpenTelemetry (Google-championed standard) for full observability
- Runs on GCP infrastructure (n2-standard-8, Cloud SQL, Cloud Run, Pub/Sub)

19. Graduated Autonomy Roadmap

Phase	Timeline	Scope	Target Metric
1: Observer (Hackathon MVP)	Weeks 1-2	All agents monitor only. Health Agent polls M_* views, RAG Agent ingests SAP docs, Browser-Use Agent extracts EWA reports and Patch Day bulletins. Zero autonomous execution.	Demonstrate architecture + X-Fix reports
2: Advisor	Month 1-2	Read-only operations autonomous. Write operations generate X-Fix proposals for human review. Risk Budget tracking active but advisory.	Human approval rate >80%
3: Operator	Month 3-6	Low/medium-risk actions execute autonomously (risk score <= 10). High-risk needs human approval. Chaos engineering tests validate reliability.	MTTR <15min, 40% auto-resolution
4: Trusted Operator	Month 6-12	Expanded scope based on trust multiplier. Most operations automated except critical (risk >= 17). Incident replay drives improvement.	MTTR <8min, 70% auto-resolution
5: Autonomous	12+ months	Full autonomous operation within policy boundaries. Human oversight shifts from approval to audit. Emergency kill switch maintained.	MTTR <3min, 85% auto-resolution

Success Metrics

Metric	Phase 1	Phase 3	Phase 5
MTTR	Baseline	<15 min	<3 min
Autonomous resolution rate	0%	40%	85%
False positive rate	<20%	<5%	<2%
Rollback rate	N/A	<10%	<3%
Basis team toil reduction	0%	40%	70%
X-Fix approval rate	N/A	>80%	>95%
Risk budget utilization	N/A	60%	85%
Patch Day assessment time	Manual	<4 hours	<1 hour

20. Risk Mitigation Strategy

Risk	Likelihood	Impact	Mitigation
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Agent executes harmful action on production HANA	Low	Severe	Policy Engine + Risk Budget + Action Certificates + mandatory pre/post checks + automatic rollback
LLM hallucination leads to incorrect diagnosis	Medium	Major	RAG grounding in SAP documentation, confidence thresholds, human review for high-risk actions
Browser-Use Agent credential exposure via prompt injection	Low	Severe	allowed_domains lockdown, sensitive_data isolation, domain-scoped tools
Risk Budget gaming (agent underreports risk)	Low	Major	Risk scores derived from operation type (not agent self-assessment), independent policy engine scoring
SAP HANA version incompatibility	Medium	Moderate	Version-aware agent configs, HANA Express testing, graceful degradation
EU AI Act non-compliance	Medium	Major	X-Fix explainability, full audit trails, human oversight by design, graduated autonomy
Vendor lock-in (Google Cloud)	Low	Moderate	OpenTelemetry, OPA (CNCF), LangGraph (MIT-licensed), containerized services

Sources

- [1] SAP Support. SAP Security Patch Day, January 2026. <https://support.sap.com/en/my-support/knowledge-base/security-notes-news/january-2026.html>
- [2] Onapsis Research Labs. SAP Security Patch Day for January 2026. <https://onapsis.com/blog/patch-day-january-2026/>
- [3] SecurityWeek. SAP's January 2026 Security Updates Patch Critical Vulnerabilities. <https://www.securityweek.com/saps-january-2026-security-updates-patch-critical-vulnerabilities/>
- [4] Pathlock. SAP Security Patch Tuesday January 2026. <https://pathlock.com/blog/security-alerts/sap-security-patch-tuesday-january-2026/>
- [5] SecurityBridge. SAP Security Patch Day, January 2026. <https://securitybridge.com/blog/sap-security-patch-day-january-2026/>
- [6] RedRays. SAP Security Patch Day January 2026. <https://redrays.io/blog/sap-security-patch-day-january-2026/>
- [7] CyberPress. SAP Security Patch Day January 2026 Addresses Critical Injection and RCE Vulnerabilities. <https://cyberpress.org/sap-injection-and-rce-vulnerabilities/>
- [8] Google Cloud. Vertex AI Agent Builder. <https://cloud.google.com/products/agent-builder>
- [9] Google Cloud. Pub/Sub Overview. <https://cloud.google.com/pubsub/docs/overview>
- [10] Open Policy Agent. <https://github.com/open-policy-agent/opa>
- [11] Google SRE Book. Embracing Risk. <https://sre.google/sre-book/embracing-risk/>
- [12] LangGraph Documentation. State Management and Agent Orchestration. <https://langchain-ai.github.io/langgraph/>
- [13] browser-use. Browser Automation Framework. <https://github.com/browser-use/browser-use>
- [14] SAP Community. Future of SAP Landscape Management. <https://community.sap.com/t5/technology-blog-posts-by-sap/future-of-sap-landscape-management/ba-p/13577450>
- [15] Avantra. SAP Operations Automation. <https://www.avantra.com/>
- [16] Permit.io. Human-in-the-Loop for AI Agents: Best Practices. <https://www.permit.io/blog/human-in-the-loop-for-ai-agents-best-practices-frameworks-use-cases-and-demo>
- [17] Arion Research. The Agentic Service Bus: Inter-Agent Communication. <https://www.arionresearch.com/blog/the-agentic-service-bus-a-new-architecture-for-inter-agent-communication>
- [18] Knight First Amendment Institute. Levels of Autonomy for AI Agents. <https://knightcolumbia.org/content/levels-of-autonomy-for-ai-agents-1>